FORGED BY FREEDOM

SLU-PP-332

SLU-PP-332 — ERRα/γ Agonist (Exercise Mimetic)

EXERCISE MIMETIC | **MITOCHONDRIAL** | **ENDURANCE** | **FAT LOSS** | **EMERGING**

WHAT IS IT?

SLU-PP-332 is a synthetic agonist of estrogen-related receptors alpha and gamma (ERRα and ERRγ) — nuclear receptors that regulate the expression of genes controlling mitochondrial biogenesis, oxidative metabolism, and energy expenditure. Published in the Journal of Medicinal Chemistry (2023), SLU-PP-332 increased endurance in mice by 70% and promoted significant fat loss independent of diet or exercise. Among the most compelling emerging exercise mimetics in preclinical research.

DOSING AT A GLANCE

No human trials yet	Preclinical dose	Extrapolated human	Route (unclear)
N/A (human)	Mouse: 30 mg/kg	~20–50 mg?	Daily SC/oral

KEY BENEFITS

- 70% increase in treadmill endurance capacity in preclinical studies (without training)
- Significant fat loss — promotes oxidative fat metabolism in skeletal muscle
- Mitochondrial biogenesis — increases mitochondrial density in muscle tissue
- Shifts muscle fiber type toward slow-twitch / oxidative (endurance-type)
- Reduces cardiac hypertrophy and improves heart failure markers in animal models
- Exercise mimetic — produces aerobic training adaptations without exercise
- Potential application in obesity, type 2 diabetes, heart failure, sarcopenia
- Synergistic with actual exercise training

HOW IT WORKS

SLU-PP-332 activates ERRα and ERRγ — orphan nuclear receptors that normally respond to energy stress signals (like those generated by exercise). Activation of these receptors triggers expression of PGC-1α target genes — the master regulator of mitochondrial biogenesis. This results in increased mitochondrial number, enhanced fatty acid oxidation, muscle fiber type shifting, and improved cardiac function, all mimicking the cellular adaptations of sustained aerobic training.

PROTOCOLS

- **IMPORTANT:** No established human protocol — all current use is self-experimentation.
- **Extrapolated from mice:** ~20–50 mg daily oral or SC (highly uncertain).
- **Stack context:** Often grouped with MOTS-c, BAM-15, 5-Amino-1MQ for metabolic/exercise mimetic protocols.
- **Monitoring:** Track cardiovascular markers, liver enzymes, and body composition changes.

SIDE EFFECTS

- Unknown in humans — no human clinical trial data available
- Preclinical: no significant toxicity observed at study doses
- Theoretical: ERR activation in cancer cells could promote proliferation (monitor)
- Unknown cardiovascular, renal, and hepatic effects in humans long-term

STORAGE

Powder: room temperature, sealed, away from moisture and light. Long-term: -20°C freezer.

■ ***Extreme caution — zero human clinical trial data. Among the most experimental compounds on this list. Use only with thorough understanding of the risk profile.***